Experiment 17: Medical Expert System

Aim:

To develop a rule-based medical expert system that infers possible diseases from patient symptoms.

Algorithm:

1. Collect symptoms from the user.
2. Match symptoms against disease rules.
3. Compute confidence or list matching diseases.
4. Present the most likely diagnosis.

Code:

```
def diagnose(symptoms):
    rules = {
        'Common Cold': {'cough', 'sneeze', 'runny_nose'},
        'Flu': {'fever', 'cough', 'body_ache', 'fatigue'},
        'Migraine': {'headache', 'nausea', 'sensitivity_to_light'},
        'Covid': {'fever', 'cough', 'fatigue', 'loss_of_taste'}
    }
    results = {}
    for disease, required in rules.items():
        match = len(symptoms & required) / len(required)
        if match > 0:
            results[disease] = round(match * 100, 1)
    return sorted(results.items(), key=lambda x: -x[1])

symptoms = {'fever', 'cough', 'fatigue'}
print("Diagnosis:")
for d, conf in diagnose(symptoms):
    print(f"  {d}: {conf}%")
```

Sample Output:

```
Diagnosis:
  Flu: 75.0%
  Covid: 75.0%
  Common Cold: 33.3%
```

Result:

The expert system matched symptoms against disease rules and ranked possible diagnoses by confidence score.